Data Structures & Algorithms (CST-003) - Unit 4

Unit 4 - Trees & Graphs - notes in easy English, matched to the UTU syllabus.

Study Materials by Anas - study-materials-by-anas.vercel.app

What this unit covers: trees and graphs - hierarchical and networked data, plus the classic walks over them.

Key ideas: A **binary tree** gives each node at most two children. A **BST** (binary search tree) keeps left < root < right, so search/insert take O(height). Traversals: **inorder** of a BST prints values sorted - a very common one-line question. Height-balanced trees (AVL) keep height O(log n). A **graph** is vertices plus edges; store it as an **adjacency matrix** (n x n array, fast edge check, more memory) or **adjacency list** (array of lists, memory-friendly). **BFS** explores level by level using a queue; **DFS** dives deep using a stack/recursion - know which data structure each uses, that question returns every year.

Exam pointers: draw a BST after inserting a given sequence, give its three traversals, and run BFS/DFS on a drawn graph writing the visit order - these three cover most of the unit's marks.

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